



EVGCPL INTRANET PORTAL

Project Document • Version 6.0 • May 2026

Endpoint Registry • PO_Actual swap • Strict-Master PCC • Hierarchical Config • Versioned Builds

Portal URL	https://evgcpladmin.github.io/evgcpl-portal/
Repository	https://github.com/EVGCPLAdmin/evgcpl-portal
Current Build	v3.4.1 • build 320 • 2026-05-05 (footer-stamped on every page)
Architecture	12 page HTMLs + portal-bundle.js (821 KB) + portal.css + multi-page-bootstrap.js + /pcc/ subapp
Endpoint Registry	5 logical endpoints — main, pinReset, aiProxy, diagnostic, pcc — overridable from Config
Apps Script files	Router.gs, SafetyHandlers.gs, ScheduledReports.gs, AIChat.gs, AiProxy.gs, SheetDiagnostic.gs, PCHandlers.gs
Auth	PIN + Google OAuth + Vendor/SC modal • Role-based nav restrictions
Scale	64 sites • 333 employees • 319 vendors • 54 SCs • 288 assets
Build script	/build-portal.js — auto-bumps build, patches constants, validates, zips with semver name

1. What's New in v6 (May 2026)

Six major changes shipped since v5, all bundled in this v3.4.1 release:

- Apps Script Endpoint Registry — all /exec URLs now managed in one place via Config →  Apps Script Endpoints. localStorage-backed overrides;  Test buttons; no code changes needed to swap deployments.
- PO Table → PO_Actual — primary tab swap on the Purchase sheet (gid 1458467853). Single PO_TAB constant; 10 fetchSheet call-sites updated.
- Strict-Master PCC — WBS Codes and Activities can ONLY be picked from M_PL_1_Activities. Free typing disabled, save validates against master, stale rows flagged.
- Workplan rebuilt — per-activity schema (one row per activity, not per month): WBS Code, Nature, Activity, UoM, Qty, Start, End, Duration, % Weight, Responsibility + 4 master-link fields + timestamp.
- Hierarchical Config page — Section accordions with status pills, per-section bulk toggles (All Live / All Off), search filter, role chips. Replaces flat module table.
- Versioned builds — PORTAL_VERSION + PORTAL_BUILD + PORTAL_BUILD_AT constants stamped into the bundle on every package; footer renders v{semver} · build {n} · {date} on every page; zip filename follows the same pattern.

2. Apps Script Endpoint Registry

Five logical Apps Script endpoints are now managed centrally. Defaults are compiled into portal-bundle.js. Runtime overrides are saved to localStorage and applied via the getExec(key) accessor.

Key	Label	Used by
main	Main Backend	Most portal POSTs (DPR, Safety, Onboarding, Reports, generic appendRow/updateCell)
pinReset	PIN Reset	v2_PINReset bound to UserSecrets sheet
aiProxy	AI Proxy (Groq)	EG Assistant chat panel
diagnostic	Sheet Diagnostic	Sharing-Doctor server-side checks
pcc	PCC Handlers	saveProjectSetup, saveBOQ, saveWBS, saveWorkplan, etc.

2.1 How it works

Constants at the top of portal-bundle.js declare the registry shape:

```
const EXEC_REGISTRY_DEFAULTS = { main: { label, desc, defaultUrl }, pinReset: {...}, ... };
const EXEC_LS_KEY = 'evgcpl_exec_registry_v1';
function getExec(key) { /* localStorage overrides → defaults */ }
```

All runtime fetches use `getExec(key)` instead of hardcoded literals. Backwards-compatible `APPS_SCRIPT_URL` and `PIN_API_URL` constants still exist but resolve through the same accessor.

2.2 Updating an endpoint without code changes

- Apps Script Editor → Deploy → New deployment (or update existing) → copy the /exec URL
- Portal → Config →  Apps Script Endpoints
- Paste into the right row →  Test → if green, click  Save endpoints
- Reload — the new URL is now live everywhere

2.3 Test mechanics

The Test button POSTs { action: '___ping___' } and surfaces response time + HTTP status as a colored pill. Even "Unknown POST action: ___ping___" is treated as green — it proves the deployment is alive. Only network errors and HTTP 4xx/5xx are red.

3. PO Table → PO_Actual

The primary Purchase Order tab on the Purchase sheet was migrated from 'PO' to 'PO_Actual' (gid 1458467853). The legacy 'PO' tab remains read-only on the same spreadsheet for archive.

3.1 Single source of truth

```
const PO_SHEET_ID = '1zcqF2tjjBETPuW25c9MBMo0zakBIBD6tksg50stFA7c';
const PO_TAB      = 'PO_Actual'; // gid 1458467853 – replaces legacy 'PO' tab as of v3.4.0
```

All fetchSheet calls now reference PO_TAB. To roll back, change one constant — PO_TAB = 'PO' — and the entire portal reverts.

3.2 Updated call sites (10 total)

- SCM dashboard — main grid, KPI strip, recent POs
- Procurement page — PO listing with status filters
- Reports module — PO trend report and PO drill-down
- Notifications scanner — pending PO approval alerts
- Vendor scorecard — PO history per vendor
- Site dashboard — PO totals per site
- Plus 4 secondary readers (analytics, exports, dashboards)

3.3 Sheets registry entry updated

The internal Sheets registry (used by the Config Sheets viewer) was updated:

```
tabs: ['MRS', 'PO_Actual', 'SiteMaster', 'VendorMaster']
```

with notes documenting the gid for future reference.

4. Strict-Master Mode for PCC

Project Cost Control's WBS step now enforces strict master compliance. Free-typing is disabled across the entire flow. WBS Codes and Activities can only be added via picker modals that read DISTINCT values from M_PL_1_Activities.

4.1 M_PL_1_Activities — the master

This is the org-wide source of truth for all activity-level data, located on the PCC backing sheet (1dQow9nD...). Schema (existing):

Checksum • Task Code • UUID • WBS Code • Project Code • Sequence # • Activity • Nature of Work • Type of Work • Unit • Site Name • UserEmail • SystemEmail • Timestamp

Three derivations from this master power the strict-mode UI:

- WBS Codes — DISTINCT(WBS Code) populates the WBS picker
- Nature of Work — for each WBS Code, the most-frequent paired Nature is auto-selected
- Type of Work — for each Nature, DISTINCT(Type of Work | Nature) populates the dependent dropdown when picking activities

4.2 What's locked vs editable

Layer	Field	Editable?
WBS row	WBS Code	 Locked — picker only
WBS row	WBS Name	 Locked — comes from master
WBS row	Nature of Work	 Locked — auto-derived
Activity row	Activity	 Locked — picker only
Activity row	WBS Code	 Locked — from master
Activity row	Nature	 Locked — inherited from WBS
Activity row	Type of Work	 Locked — from master
Activity row	Cost Code	Editable — project-specific
Activity row	Unit	 Locked — from master
Activity row	BOQ Qty	Editable — project-specific

4.3 Save validation

Before submitting, save() runs strict checks:

- Every WBS Code is verified against validWbsCodes (Set built from master at load time)
- Every Activity is verified against validChecksums

- If any row fails, save is BLOCKED with an alert listing the offending entries — user must remove or re-pick before proceeding

4.4 Stale row handling

If pre-existing data references master rows that no longer exist (or never did), they load with:

- Gold/red border +  stale tag in the UI
- Status pill at top of page:  N stale rows
- Save blocked until cleaned up

5. Workplan — Per-Activity Schema

Workplan was rewritten with a one-row-per-activity schema. The previous monthly-grid model (one row per activity per month) was retired in favor of a more flexible Start/End/Duration/Weight model that better fits how activities are actually scheduled.

5.1 Workplan tab — 15 columns

Column	Type	Source / Notes
Project Code	string	Active project context
WBS Code	string	Inherited from activity's WBS
Nature of Work	string	Inherited from WBS's Nature
Activity	string	From master via CheckSum
UoM	string	From master • Unit
Qty	number	BOQ Qty • editable here
Start	YYYY-MM-DD	User-entered
End	YYYY-MM-DD	User-entered
Duration	integer (days)	Auto = (End – Start) + 1
% Weight	number 0–100	Target sum across project = 100
Responsibility	string	Employee Register dropdown
Master UUID	string (FK)	→ M_PL_1_Activities.UUID
Task Code	string	Denormalized from master
CheckSum	string (FK)	→ M_PL_1_Activities.CheckSum
Updated At	ISO timestamp	Server-stamped on every save

5.2 KPIs

- Activities — count linked from WBS
- Total Weight — sum of % Weight, color-coded: green at 100, gold below, red above
- Project Span — earliest Start → latest End across all rows
- Total Duration — sum of activity durations

5.3 Behavior

- Duration auto-computes from Start..End on every cell change
- Responsibility dropdown sourced live from the Employee Register sheet (falls back to free text if unreachable)

- Save shows a confirm prompt if weights don't sum to ~100% (warning, not block)
- '↺ Sync from WBS' re-pulls the activity list from the latest WBS state

6. Hierarchical Config Page

The flat module-and-roles table was replaced with a Section → Module accordion. Each section collapses into a single row showing aggregate state; expanding reveals per-module rows.

6.1 Section accordion

Each of the 11 sections (Main • HR & People • Site Ops • Procurement • Finance • Planning • Plant & Machinery • Reports • Quick Access • Personal • Admin) shows:

- ► chevron + section name (click anywhere to expand)
- State pill: All Live (green) / All Off (grey) / Mixed (amber)
- Module count + per-status chips: '3 live • 1 dev • 1 off'
- Role grants count: '21/35 role grants' for at-a-glance permission density
- Per-section bulk: 'All Live' / 'All Off' buttons that flip the entire section

6.2 Top toolbar

-  Live search — type 'dpr' or 'payment' and only matching modules show; matching sections auto-expand
- Expand all / Collapse all
- Global bulk: All Live / All Off (with confirm)

6.3 Module rows

When expanded, each module is a horizontal grid row with:

- Label + route code (e.g. /budgeting in mono font)
- Status dropdown — color-coded with matching pill background
- Role chips — pill toggles for each role; first-word fits ('MD', 'HR', 'Site'...) with tooltip showing full label
- ↗ Open button — jump to the module immediately

6.4 Self-heal logic

CONFIG_KEY was bumped from evgcpl_portal_config to evgcpl_portal_config_v2. loadPortalConfig now backfills missing routes from MODULE_REGISTRY defaults, so newly-added modules (like Budgeting + Execution) are never silently invisible after a registry update.

7. Versioning & Build Process

7.1 Constants

Three version constants live at the top of portal-bundle.js:

```
const PORTAL_VERSION = '3.4.1';
const PORTAL_BUILD   = 320;
const PORTAL_BUILD_AT = '2026-05-05T18:30:00Z';
```

7.2 Footer

multi-page-bootstrap.js auto-injects #portalVersionFooter on every page load, displaying:

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Fixed-bottom strip with backdrop blur, pointer-events pass through except on the text. Mobile (<600px) drops the date and tail.

7.3 Build script

Run from the repo root:

```
node build-portal.js           // auto-bump build only
node build-portal.js --patch   // 3.4.1 → 3.4.2, build++
node build-portal.js --minor    // 3.4.1 → 3.5.0, build++
node build-portal.js --major    // 3.4.1 → 4.0.0, build++
node build-portal.js --version=4.0.0 // explicit semver, build++
```

The script:

- Reads version.json
- Applies the bump and updates the build timestamp
- Patches the three constants in portal-bundle.js
- Runs node --check to validate JS syntax
- Saves the new version.json
- Creates EVGCPL_Portal_v{semver}_build{n}_{YYYYMMDD}.zip

7.4 Bump conventions

Trigger	Bump
Bug fix or small UX tweak	--patch (3.4.0 → 3.4.1)
New module / route / section ships	--minor (3.4.x → 3.5.0)
Architecture change (e.g. v2 single-file → v3 multipage)	--major (3.x.x → 4.0.0)
Every package, regardless of bump	build++ (always increases monotonically)

8. Apps Script Backend

All server-side logic lives in a Google Apps Script project bound to the master spreadsheet, deployed as a Web App at the exec URL. Files in /apps-script/ in the repo are reference copies — they need to be manually pasted into the Apps Script Editor and a new version deployed for changes to be live.

8.1 File map

File	Purpose
Router.gs	doPost / doGet entry points + action dispatch table. Every route is documented here.
SafetyHandlers.gs	Generic write API: appendRow, updateCell, batchUpdate. Plus closeSafetyIncident.
ScheduledReports.gs	Time-driven email triggers; saveScheduledReport, runReportNow.
AIChat.gs	aiChat action — Groq llama-3.3-70b-versatile via Apps Script proxy.
AIProxy.gs	Generic LLM proxy with Gemini fallback (uses GEMINI_API_KEY if Groq unavailable).
SheetDiagnostic.gs	diagnoseSheet, listShares — server-side sheet sharing checks.
PCCHandlers.gs	10 PCC actions: saveProjectSetup, saveBOQ, saveWBS, saveWorkplan, saveManpower, saveMachinery, saveMaterials, saveOverheads, saveVariations, submitBudgetApproval.
README.md	Install/deploy instructions + full action map + common gotchas.

8.2 Deploy cycle

After editing any .gs file:

- Save in the Apps Script Editor
- Deploy → Manage deployments →  on the active deployment
- Version: New version → Deploy
- The exec URL stays the same. The new code is live within seconds.

Editing a .gs file alone is NOT enough — Apps Script Web Apps cache the deployed snapshot. Redeploying a new version is mandatory.

8.3 Required Script Properties

- GROQ_API_KEY — gsk_... key from console.groq.com (free tier: 30 req/min)
- GEMINI_API_KEY — optional fallback for AI when Groq is down

9. Sheet IDs & Constants

Constant	Sheet ID	Purpose
EXEC_REGISTRY NEW	5 logical Apps Script endpoints	Runtime overrides via Config
SHEET_ID (Master)	1B2wb38KhNwlLoZnsAGWQkO0FdEGFFfsh3ycRRurigq4	BillingMaster, SiteMaster
EMP_SHEET_ID	1HWKZPhKRhcuVxBgyyN8zRt8p-SzYmKjJWiOdCgykBHs	Employee Register
V2_MASTER_SHEET_ID	1fhSO4WBYP0LNXPxe9I9zr5qsIPs9CIDFpUixBogPnsM	v2 Master
DPR_SHEET_ID	139deMPqCXVZLSw5gFdzhN1heV-hynkYqljsVpO1oXSc	Daily Progress Reports
PO_SHEET_ID	1zcqF2tjjBETPuW25c9MBMo0zakBIBD6tksg5OstFA7c	Purchase / PO
PO_TAB NEW	'PO_Actual' (gid 1458467853)	Replaces legacy 'PO' tab
PAYMENT_SHEET_ID	1mLddxLRf719EaXE9XSET9gT8lOa8Cxns362yIbHo63g	Payments
STORES_SHEET_ID	1iMQxgqGilUh2_3NCZl5D-EMt-NC8FwugX83q2fWb8fE	Stores • StockIN • GRN
SAFETY_SHEET_ID	1B8P0PawV43ksazbzhKsil1X6-INOfxx9PFvGycNOvDY	Safety • Incidents
REWARDS_SHEET_ID	1vz8HLopjISF8TF7rzYuVu5JjqukT929I7a5x7kdehll	Rewards & Wall
PIN (UserSecrets)	1hN4VEDNpVLD3lKuBPYCTOaViv7UpveRfud2d2gy15D0	UserSecrets
PCC backing sheet	1dQow9nD4e0qVOSfpwEWQmPTuhF3FW_8r1oK5dMjJIRE	PCC + M_PL_1_Activities
BUDGET_SHEET_ID	(empty — pending Drive upload)	IC Budget template

10. Open Points & Pending Actions

10.1 User actions required

#	Item	Notes
1	Deploy Apps Script Router.gs	Paste Router.gs into the bound Apps Script project to wire the 30+ actions. Without this, PCC saves return 'Unknown POST action'.
2	Paste PCCHandlers.gs	Sibling file to Router.gs that contains the 10 PCC handler functions
3	Redeploy Web App	Deploy → Manage deployments →  → New version → Deploy. Exec URL stays the same.
4	Confirm PO_Actual sharing	Sheet 1zcgF2tjj.../gid 1458467853 must be shared 'Anyone with the link' for portal reads
5	Verify GROQ_API_KEY	Apps Script → Project Settings → Script Properties. AI Assistant is unresponsive without this.
6	Test all endpoints after deploy	Portal → Config →  Apps Script Endpoints →  Test all (should be 5 green pills)
7	Upload IC Budget Excel	Set BUDGET_SHEET_ID for the Afcons-format IC Budget template (separate from PCC)

10.2 Dev pending

- Grade_MatKey composite column in 10_Mix_Design_Lookup (HIGH)
- AppSheet CSV export with Virtual Columns (MED — awaiting clarification on target app)
- My Team reportees card debug — empCode vs managerCode value mismatch (MED)
- Onboarding checklist — final item list pending HR confirmation (MED)
- Payslip PDF generation — phase 3 payroll integration (LOW)
- SC Work Orders / Job Cards — sheet not yet available (LOW)

10.3 Architecture / infrastructure

- CNAME: intranet.evgcpl.com → not planned for now
- OAuth External audience → not required (PIN + Google + Vendor modal cover all)
- Repo URL update → 'evgcpladmin' → 'EVCPLAdmin' (use git remote set-url)

11. Build Log

Version	Date	Key Changes
v1.0	Jan 2026	Initial portal — Dashboard, HR, Accounts, SCM. Sidebar nav. PIN auth. gviz fetch.
v2.0	Feb 2026	Top nav bar. Rewards & Wall. AppSheet/Sheets split. Safety. Site Manager.
v3.0	Mar 2026	Onboarding. My Profile. My Team. Scheduled emails. Planning skeleton.
v4.0	Apr 2026	My Profile redesign. Per-report scheduling. Apps/Sheets split. Planning iframe.
v5.0	May 2026	PCC subapp integrated. Planning → Budgeting + Execution. Dashboard + Command Center merge for MD. DPR sheet wired.
v6.0	May 2026	Apps Script Endpoint Registry • PO → PO_Actual swap • Strict-Master PCC (WBS + Activities picker only) • Workplan rebuilt (per-activity schema) • Hierarchical Config page • Versioned builds with footer stamping. Bundle: v3.4.1 build 320.

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